

Fuels : Classification of fuels, their merits and limitations. (2 L)

Solid Fuels: Origin of coal, types, properties, proximate and ultimate analysis, storage and reserves in India. Coal washing, preparation and blending methods. Applications of coal.

Coke making by beehive and by-product ovens. Modern practices of coke making. Principles of graphitisation and reactivity. Characterization of coal and coke. (9 L)

Selection of reductant /fuel for Blast Furnace, DRI, COREX, Cupola and Pit Furnace. (15 L)

Liquid and Gaseous Fuels: Types and uses of liquid and gaseous fuels. Flame characteristics. Burners for liquid, gas and pulverized coal. Synthesis and reformation of gas for direct reduction, Producer gas and Water gas. (6 L)

Refractories: Classification of refractories. Properties and applications of Fireclay, Silica, Chromite, Carbon/Graphite, Magnesite, Dolomite, Zirconia, Silicon Carbide, Sillimanite, Kyanite refractories. (6 L)

Selection of Refractories: Blast Furnace, LD Converter, Electric Arc Furnace. (1 L)

4. READINGS

4.1 TEXTBOOKS::

1. O.P. Gupta: Elements of Fuels, Furnaces and Refractories, Khanna Publishers (Delhi).
2. Efficient Use of Fuel, HMSO (London).
3. J.D. Gilchrist: Fuels, Furnaces and Refractories, Pergamon.
4. RC Gupta : Theory and laboratory experiments in ferrous metallurgy, PHI, New Delhi, 2010

4.2 *REFERENCE BOOKS::

1. D.N. Nandi: Handbook of Refractories, Tata McGraw-Hill.
2. J.H. Chestors: Steel Plant Refractories; United Steel Companies, Sheffield.
3. F.H. Norton: Refractories, McGraw-Hill.